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The Systems Behind Modern AI

Abstract

Modern machine learning and AI runs primarily on Graphics Processing Units (GPUs), hardware initially designed to render graphics for video games. We briefly discuss the history of GPU architecture, and the motivations behind some key innovations. Because neural networks rely heavily on matrix multiplication, efficient implementations of matrix multiplication are crucial in modern AI systems. We explore these implementations through the CUDA programming model, and introduce the roofline model to reason about computational efficiency and hardware utilization. The growing demand for AI-specific compute has also led to the development of alternative parallel processing architectures including Tensor Processing Units, Neural Processing Units, and Wafer Scale Engines. We conclude by briefly considering the future evolution of these systems and the architectural motivations driving it.